FORMATIVE ASSESSMENT 1

Research assignment

Beginner Java

JD521

2024

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CTU training solutions
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Please complete the declaration of authenticity below for all assignments:

DECLARATION OF AUTHENTICITY

I _____Chane Marais_____ hereby

(FULL NAME)

declare that the contents of this assignment are entirely my own work with the exception of the following elements: (List the elements of work in this project that were not self-generated as well as who the originator of the element is)

Element	Originator

Signature: CM **Date: 10 June 2024**

Programming Foundation

Module: Beginner Java

Module Code: JD521

NQF: 5

Credits: 22

Assessment type: Formative Assessment

Mark allocation: 100

Hand out date: 08 March 2024, 08:00

Hand in date: 22 March 2024, 23:30

Instructions:

Read each question carefully and consider the mark allocation prior to answering.

- Ensure you answer all questions.
- Section A: Multiple choice questions – **40** marks.
- Section B: Short Questions – **30** marks.
- Section C: Research Questions – **30** marks.
- For final submission**,** include the Declaration of Authenticity **and your Answer Sheet (PDF Document)**

Unit Standards:

- Demonstrate an understanding of the handling of errors in a computer programming environment
- Apply the principles of designing computer system inputs and outputs
- Manage software development source files using appropriate tools
- Demonstrate logical problem solving and error detection techniques
- Demonstrate an understanding of sort and search techniques used in computer programming
- Test a computer program against a given specification.

Section A: Multiple Choice Questions (40 Marks)

Instructions: Choose the correct option for each question below.

1. Which of the following is a valid declaration of a char array in Java? (2 Marks)

   A. `char[] chars = new char(10);`
   B. `char chars[] = "Java";`
   C. `char[] chars = new char[]{'J', 'a', 'v', 'a'};`
   D. `char chars[] = new char[]{"Java"};`

2. What is the result of the following code snippet? (2 Marks)

```
double result = 10 / 4;
System.out.println(result);
```

   A. 2 B. 2.5 C. 2.0
   D. Compilation Error

3. Which of the following for-loop declarations is invalid in Java? (2 Marks)

   A. `for(int i = 0; i < 5; i++){}`
   B. `for(int i = 0; i < 5;){i++;}`

C. `for(;;){break;}`

D. `for(int i = 0, String s = "Java"; i < s.length(); i++){}`

4. What will the following switch statement print? (2 Marks)

```
int key = 2;
switch(key) {
case 1:
        System.out.println("One");
case 2:
        System.out.println("Two");
case 3:
        System.out.println("Three");
default:
        System.out.println("Default");
}
```

A. Two
B. TwoThreeDefault
C. Three
D. Default

5. Which principle of OOP is violated when 'a subclass changes the behaviour of a superclass method'? (2 Marks)

A. Encapsulation
B. Inheritance
C. Polymorphism
D. Abstraction

6. What is the output of the following code snippet, assuming all necessary imports are done and files exist? (2 Marks)

```
class Animal { } class Dog
extends Animal { } public
class Test {
    public static void main(String[] args) {
        Animal a = new Dog();
        System.out.println(a instanceof Dog);
System.out.println(a instanceof Animal);
    }
}
```

A. true true B. true false C. false true
D. false false

7. Which of the following is true about the 'finally' block in Java? (2 Marks)

A. It runs after the try block but before any catch blocks.

B. It does not execute if the program exits (either by calling System.exit() or due to a fatal error) within the try or catch block.
C. It is executed even if there is a return statement in the try or catch block.
D. It must accompany either a try block or a catch block.

8. Consider the following code snippet. What will be the output? (2 Marks)

```
try {     int a = 50/0;
} catch(ArithmeticException e) {
    System.out.println("Arithmetic Exception");
} catch(Exception e) {
    System.out.println("General Exception");
} finally {
    System.out.println("Finally block");
}
```

A. Arithmetic Exception
B. General Exception
C. Arithmetic ExceptionFinally block
D. General ExceptionFinally block

9. Which of these is not a difference between HashSet and TreeSet in Java? (2 Marks)

A. HashSet allows null elements; TreeSet does not.
B. TreeSet is implemented using a tree structure; HashSet uses a hash table.
C. TreeSet guarantees iteration in ascending order; HashSet does not guarantee any specific order.
D. HashSet is faster for operations like add, remove, and contain compared to TreeSet.

10. What will happen if you try to compile and run the following code snippet? (2 Marks)

```
List<Integer> list = new ArrayList<>();
list.add(10); list.add("Java"); // Line
X
```

A. Compilation error at Line X
B. A ClassCastException at runtime
C. The code will compile and run without issues
D. A NullPointerException at runtime

11. Which tool is used for static code analysis in Java to find bugs, vulnerabilities, and code smells? (2 Marks)

A. Maven
B. Gradle
C. SonarQube
D. JUnit

12. What is the purpose of the JVM (Java Virtual Machine)? (2 Marks)

A. To compile Java source code into executable code
B. To provide a runtime environment for executing Java bytecode
C. To improve the performance of Java applications via just-in-time compilation D. B and C are correct

13. Which keyword is used to manually throw an exception in Java? (2 Marks)

A. throw B. throws
C. try
D. finally

14. When is a try-with-resources statement useful in Java? (2 Marks)

A. When you need to manually manage the lifecycle of one or more resources.
B. When the resources used, like files or sockets, implement the AutoCloseable or Closeable interface and should be closed automatically after use.
C. When you need to handle multiple exceptions in a single catch block.
D. When you want to execute a block of code regardless of whether an exception is thrown.

15. What is true about the Comparator interface in Java? (2 Marks)

A. It is used to order the objects of user-defined classes.
B. It provides a single sort() method to sort objects.
C. It is found in the java.util.Comparator package.
D. It can only be used with objects that implement the Comparable interface.

16. Which of the following statements about generics in Java is incorrect? (2 Marks)

A. Generics add stability to your code by making more of your bugs detectable at compile time.
B. Generics allow the type (class or interface) to be a parameter to methods, classes, and interfaces.
C. With generics, it is possible to create classes that work with different data types.
D. Generics can only be used with collection classes like ArrayList and HashMap.

17. What is the function of the JDK (Java Development Kit)? (2 Marks)

A. It provides the runtime environment only.
B. It compiles Java source code into bytecode only.
C. It includes tools for developing, debugging, and monitoring Java applications.
D. It enhances the performance of Java applications on a specific platform.

18. Which of the following tools is used to manage dependencies and build Java projects? (2 Marks)

A. Docker
B. Git
C. Maven
D. Jenkins

19. In Java, which of the following is not a valid way to create a thread? (2 Marks)

A. By implementing the Runnable interface.
B. By extending the Thread class.
C. By implementing the Callable interface and submitting to an ExecutorService.
D. Directly instantiating the Thread class with a lambda expression.

20. Which statement about lambda expressions in Java is true? (2 Marks)

A. They can be used to implement any interface, regardless of the number of abstract methods.
B. They allow the function to be treated as an argument, or code as data.
C. They replace the need for anonymous inner classes.
D. They are objects and can be assigned to any variable type.

End of section A. **[Total = 40 Marks]**

Section B: Scenario Questions (30 Marks)

Instructions: Answer the following question.

2.1 Debug the following code snippet and identify the 15 errors along with the correct snippet. (30 Marks)

```
1 import java.utl.Scanner;
2
3 public classe DebuggingExercise {
4
5     public static void Main(String[] args) {
6     Scanner input = new Scanner(System.in); 7
8          Sstem.out.println("Enter an integer: ");
9          int number = input.nextint();
10
11         int result = sumNumbers(upTo: 5);
12         System.ot.println("Sum up to 5 is: " + result);
13
14    if(number = 10) {
15    System.out.println("You entered ten!"); 16        }
17
18    // Calculate factorial
19    int factorial = 1;
20    for(int i = 2; i <= number i++) {
21    factorial *= i; 22        }
23
24         System.out.println("Factorial of " + number + "is: "
   + factorial);
25
26               int divisionResult = number / 0;  27
28     System.out.println("Division result: " + divisionResult);
29     }
30
31             pubic static int sumNumbers(int upTo) {
32             int sum = 0;
33             for(int i = 0; i <= upTo; i++)
34             sum += i;
35             return sum
36             }
```

End of section B. **[Total = 30 Marks]**

Section C: Application Questions (30 Marks)

Instructions: Complete the following task, and attach your code snippet or screenshots on your answer sheet

3.1 Develop the following program. (30 Marks)

*Your program will be a **'basic'** calculator that performs arithmetic operations (addition, subtraction, multiplication, and division) on two numbers entered by the user.*

It must meet the following criteria:

Use of Imports
Utilize the 'Scanner' class from the 'java.util' package to read input from the user.

Object Handling:

- Create a Calculator class that contains methods for each arithmetic operation: add, subtract, multiply, and divide.
- Instantiate a Calculator object in your main method and use it to perform the operations based on the user input.

Exception Handling:

- Implement exception handling to catch any ArithmeticException that occurs during division (e.g., division by zero).
    - Display a friendly error message to the user in case of such an exception.
- Use a try-catch block to manage any potential InputMismatchException when reading the numbers, indicating the user should enter valid numeric values.

End of section C **[Total = 30 Marks]**

End of paper.

Question 3 Rubric

Evaluation Criterion	Mark Allocation	Mark Awarded
Use of Imports		
Scanner import	2	
Object handling		
Calculator Class created	4	
The calculator class contains methods for sum, difference, product and quotient.	8	
Instance of the calculator class used to parse and invoke.	4	
Exception Handling		
ArithmeticException implemented.	6	
Try-Catch for InputMismatchException implemented.	6	
TOTAL		**/30**

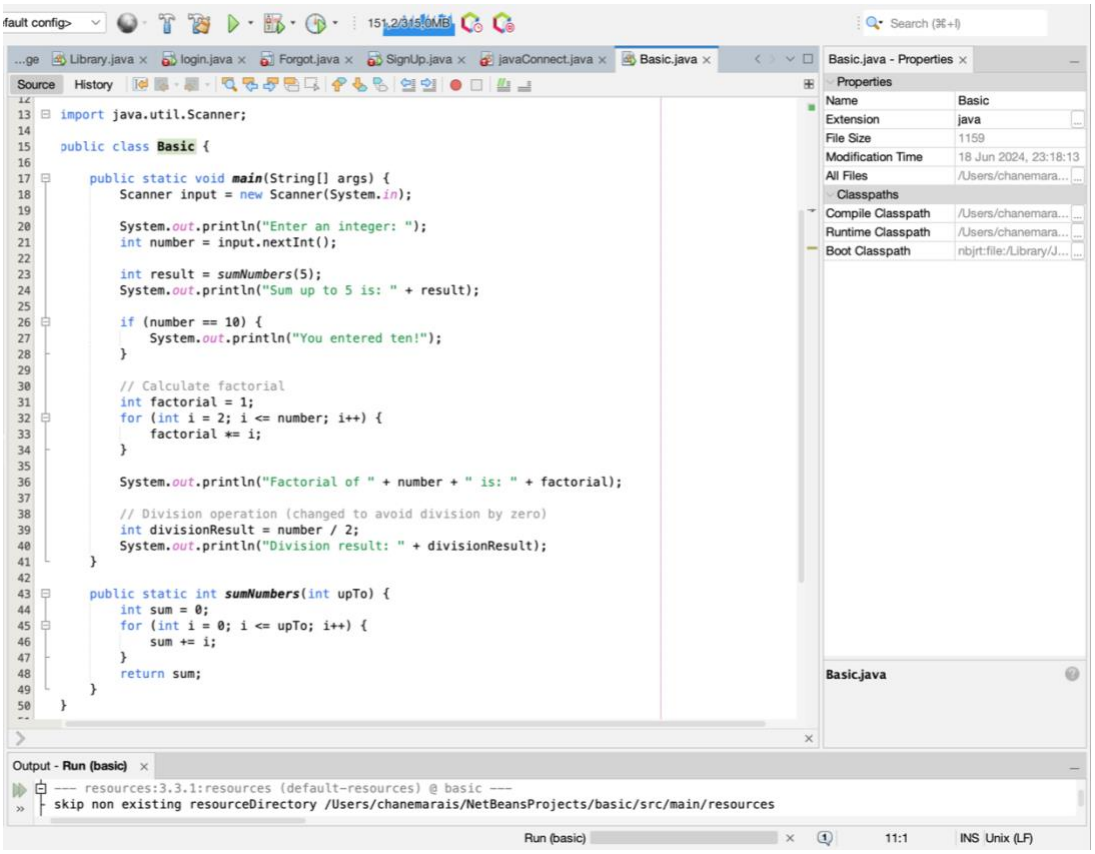
Section A: Multiple Choice Questions

1. C
2. C
3. D
4. B
5. C
6. A
7. B
8. C
9. D
10. A
11. C
12. B
13. A
14. B
15. A
16. D
17. C
18. C
19. D
20. B

Section B: Scenario Questions

Debugging code

Line 1: // Error: Incorrect package name "java.utl.Scanner"
Line 2: none
Line 3: // Error: "classe" should be "class"
Line 4: none
Line 5: // Error: "Main" should be "main"
Line 6: // Line number should not have "7" after the statement. Remove "7".
Line 7:none
Line 8: // Error: "Sstem" should be "System"
Line 9: // Error: "nextint" should be "nextInt"
Line 10:none
Line 11: // Error: Incorrect method call syntax, should not use ":" for parameter passing
Line 12: // Error: "System.ot" should be "System.out"
Line 13:none
Line 14: // Error: Should use "==" for comparison instead of "="
Line 15:none
Line 16: // Line number should not have "16" before the statement. Remove "16".
Line 17:none
Line 18:none
Line 19:none
Line 20: // Error: Missing ";" after "number" in the for loop header // Corrected:
Line 21: // Line number should not have "22" after the statement. Remove "22".
Line 22:none
Line 23:none
Line 24: // Error: Missing space before "is:" in the println statement
Line 25:none
Line 26: // Error: Division by zero is not allowed, should throw ArithmeticException
Line 27:none
Line 28:none
Line 29:none
Line 30:none
Line 31: // Error: "pubic" should be "public"
Line 32:none
Line 33:none
Line 34:none
Line 35: // Error: Missing ";" after "sum" in the return statement
Line 36:none

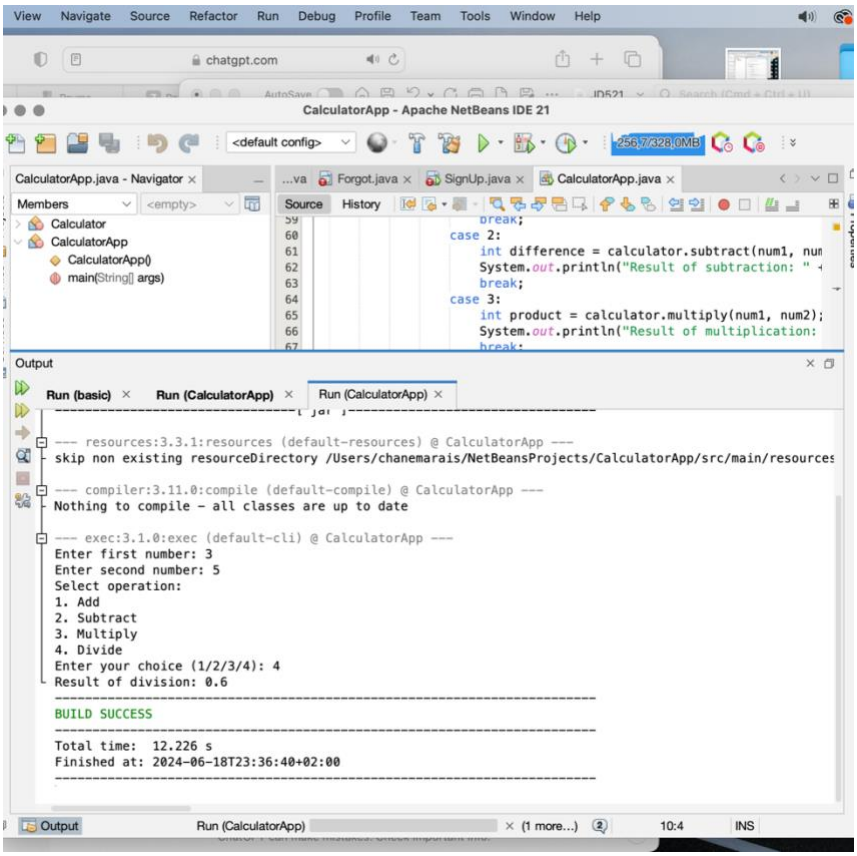



```java
import java.util.Scanner;

public class Basic {

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.println("Enter an integer: ");
        int number = input.nextInt();

        int result = sumNumbers(5);
        System.out.println("Sum up to 5 is: " + result);

        if (number == 10) {
            System.out.println("You entered ten!");
        }

        // Calculate factorial
        int factorial = 1;
        for (int i = 2; i <= number; i++) {
            factorial *= i;
        }

        System.out.println("Factorial of " + number + " is: " + factorial);

        // Division operation (changed to avoid division by zero)
        int divisionResult = number / 2;
        System.out.println("Division result: " + divisionResult);
    }

    public static int sumNumbers(int upTo) {
        int sum = 0;
        for (int i = 0; i <= upTo; i++) {
            sum += i;
        }
        return sum;
    }
}
```

Section C: Application Questions

```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 */

package com.mycompany.calculatorapp;

/**
 *
 * @author chanemarais
 */
import java.util.InputMismatchException;
import java.util.Scanner;

class Calculator {
    public int add(int a, int b) {
        return a + b;
    }

    public int subtract(int a, int b) {
        return a - b;
    }

    public int multiply(int a, int b) {
        return a * b;
    }

    public double divide(int a, int b) {
        if (b == 0) {
            throw new ArithmeticException("Division by zero is not allowed");
        }
        return (double) a / b;
    }
}

public class CalculatorApp {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        Calculator calculator = new Calculator();

        try {
            System.out.print("Enter first number: ");
            int num1 = scanner.nextInt();

            System.out.print("Enter second number: ");
```

```java
            System.out.print("Enter second number: ");
            int num2 = scanner.nextInt();

            System.out.println("Select operation:");
            System.out.println("1. Add");
            System.out.println("2. Subtract");
            System.out.println("3. Multiply");
            System.out.println("4. Divide");
            System.out.print("Enter your choice (1/2/3/4): ");
            int choice = scanner.nextInt();

            switch (choice) {
                case 1:
                    int sum = calculator.add(num1, num2);
                    System.out.println("Result of addition: " + sum);
                    break;
                case 2:
                    int difference = calculator.subtract(num1, num2);
                    System.out.println("Result of subtraction: " + difference);
                    break;
                case 3:
                    int product = calculator.multiply(num1, num2);
                    System.out.println("Result of multiplication: " + product);
                    break;
                case 4:
                    try {
                        double quotient = calculator.divide(num1, num2);
                        System.out.println("Result of division: " + quotient);
                    } catch (ArithmeticException e) {
                        System.out.println("Error: " + e.getMessage());
                    }
                    break;
                default:
                    System.out.println("Invalid choice. Please enter a number from 1 to 4.");
                    break;
            }
        } catch (InputMismatchException e) {
            System.out.println("Invalid input. Please enter valid numeric values.");
        } finally {
            scanner.close();
        }
    }
}
```



Code:

```
*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 */

package com.mycompany.calculatorapp;

/**
 *
 * @author chanemarais
 */
import java.util.InputMismatchException;
import java.util.Scanner;

class Calculator {
   public int add(int a, int b) {
      return a + b;
   }

   public int subtract(int a, int b) {
      return a - b;
   }

   public int multiply(int a, int b) {
```

```
        return a * b;
    }

    public double divide(int a, int b) {
        if (b == 0) {
            throw new ArithmeticException("Division by zero is not allowed");
        }
        return (double) a / b;
    }
}

public class CalculatorApp {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        Calculator calculator = new Calculator();

        try {
            System.out.print("Enter first number: ");
            int num1 = scanner.nextInt();

            System.out.print("Enter second number: ");
            int num2 = scanner.nextInt();

            System.out.println("Select operation:");
            System.out.println("1. Add");
            System.out.println("2. Subtract");
            System.out.println("3. Multiply");
            System.out.println("4. Divide");
            System.out.print("Enter your choice (1/2/3/4): ");
            int choice = scanner.nextInt();

            switch (choice) {
                case 1:
                    int sum = calculator.add(num1, num2);
                    System.out.println("Result of addition: " + sum);
                    break;
                case 2:
                    int difference = calculator.subtract(num1, num2);
                    System.out.println("Result of subtraction: " + difference);
                    break;
                case 3:
                    int product = calculator.multiply(num1, num2);
                    System.out.println("Result of multiplication: " + product);
                    break;
                case 4:
                    try {
                        double quotient = calculator.divide(num1, num2);
                        System.out.println("Result of division: " + quotient);
                    } catch (ArithmeticException e) {
                        System.out.println("Error: " + e.getMessage());
                    }
                    break;
                default:
                    System.out.println("Invalid choice. Please enter a number from 1 to 4.");
```

```
                break;
            }
        } catch (InputMismatchException e) {
            System.out.println("Invalid input. Please enter valid numeric values.");
        } finally {
            scanner.close();
        }
    }
}
```